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EXPERIMENT 13

Minimax Algorithm

Aim

To implement the **Minimax Algorithm** for a two-player gaming problem using Python.

Algorithm

1. Start the game with an initial state.
2. Generate all possible moves from the current state.
3. If the game reaches a terminal state, return its score.
4. For the **Max player**, select the maximum score.
5. For the **Min player**, select the minimum score.
6. Continue recursively until the best move is found.
7. Display the optimal move and final result.

Program

```
def minimax(depth, nodeIndex, maximizingPlayer, values, height):
    # Terminal node
    if depth == height:
        return values[nodeIndex]


    if maximizingPlayer:
        return max(
            minimax(depth + 1, nodeIndex * 2, False, values, height),
            minimax(depth + 1, nodeIndex * 2 + 1, False, values, height)
        )
    else:
        return min(
            minimax(depth + 1, nodeIndex * 2, True, values, height),
            minimax(depth + 1, nodeIndex * 2 + 1, True, values, height)
```

```
    )

# Input
values = [3, 5, 2, 9, 12, 5, 23, 23]

height = 3

# Apply Minimax
best_score = minimax(0, 0, True, values, height)

print("Game values:", values)
print("Best score using Minimax:", best_score)
```

Input

Game values:

3 5 2 9 12 5 23 23

Output

```
Game values: [3, 5, 2, 9, 12, 5, 23, 23]
Best score using Minimax: 12
```

Result

Thus, the Minimax Algorithm was successfully implemented in Python, and the optimal game score was obtained as 12.